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Midterm Examination - 2

Phys332: Electromagnetic Theory II

2026/04/15

Please carefully read below before proceeding!

I acknowledge by taking this examination that I am aware of all academic honesty conducts that govern this course and how they also apply for this examination. I therefore accept that I will not engage in any form of academic dishonesty including but not limited to cheating or plagiarism. I waive any right to a future claim as to have not been informed in these matters because I have read the syllabus along with the academic integrity information presented therein.

I also understand and agree with the following conditions:

(1) all calculations are to be conducted in the notations and conventions of the formulae sheets provided during the exam unless explicitly stated otherwise in the question;

(2) I take *full responsibility* for any ambiguity in my selections in "multiple choice questions";

(3) incorrect selections will receive $-1/7$ of the question's points;

(4) I am expected to provide *step-by-step explanation of how I solved the question* and am expected to do so *only within the answer boxes* provided with the questions: the explanation is supposed to be succinct, well-articulated, and correct both scientifically and mathematically;

(5) no partial credit is awarded for the explanations provided in the answer boxes;

(6) some questions of some students will be randomly selected for inspection: *a question (if selected for inspection) might be awarded negative points* if its explanation is incorrect or insufficient to get the correct answer, even if the correct option is selected;

(7) any page which does not contain *both my name and student id* may not be graded;

(8) any extra sheet that I may use are for my own calculations and will not be graded.

Signature: _____

Question:	1	2	3	4	Total
Points:	31½	31½	21	21	105

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All dimensionfull quantities are to be understood in SI units.

(For instance, $\epsilon_0 \simeq 10^{-11}$ and $|\mathbf{B}| = 5$ stand for $\epsilon_0 \simeq 10^{-11}\ \mathrm{A}^2\ \mathrm{kg}^{-1}\ \mathrm{m}^{-3}\ \mathrm{s}^4$ and $|\mathbf{B}| = 5\ \mathrm{A}^{-1}\ \mathrm{kg}^1\ \mathrm{s}^{-2}$.)

Question: 1: Eddy currents ..... *(31½ points)*
(Midterm One Question Reloaded)

For a rigid rotating object of angular velocity $\boldsymbol{\omega}$, linear velocity is given as $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$.
Please take $\epsilon_0 \simeq 10^{-11}$ and $\mu_0 \simeq 10^{-6}$ for this question.
One can derive from the Maxwell equations (see the formulae sheet) the continuity equation $\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0$.
Please take Ohm's law as $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.

Imagine a uniform, electrically neutral, spherical shell, of inner radius R_1, outer radius R_2, and conductivity σ, centered at the origin and spinning with constant angular velocity $\boldsymbol{\omega}$, in the presence of a uniform magnetic field $\mathbf{B}$. We will compute the eddy currents in the shell, in the steady state (fields, charges, currents all constant in time).

(a) **(10½ points)** Which one below would give the right expression for charge density ρ? *Hint: combine Ohm's law, Gauss law, and continuity equation.*

■ $\rho = -\epsilon_0 \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = \epsilon_0 \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = -\mu_0 \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = \mu_0 \nabla \cdot (\mathbf{v} \times \mathbf{B})$

☐ $\rho = -\epsilon_0^{-1} \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = \epsilon_0^{-1} \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = -\mu_0^{-1} \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = \mu_0^{-1} \nabla \cdot (\mathbf{v} \times \mathbf{B})$

(b) **(10½ points)** For $\mathbf{B} = -5\hat{z}$ and $\boldsymbol{\omega} = 10\hat{z}$, what would be ρ?

☐ $\rho = -10^{-9}$ ■ $\rho = 10^{-9}$ ☐ $\rho = -5 \times 10^{-10}$ ☐ $\rho = 5 \times 10^{-10}$

☐ $\rho = -10^{13}$ ☐ $\rho = 10^{13}$ ☐ $\rho = -5 \times 10^{12}$ ☐ $\rho = 5 \times 10^{12}$

(c) **(10½ points)** Consider two points, point A on the outer surface and point B on the inner surface, and assume that the origin is part of the line segment $|AB|$. What would be the ratio $\left(\frac{\partial V}{\partial r}\Big|_{\text{at the point } B}\right) / \left(\frac{\partial V}{\partial r}\Big|_{\text{at the point } A}\right)$ for the electric potential V and the radial distance r? *Hint: combine Ohm's law along with the fact that eddy currents here have no radial components.*

☐ $-\sqrt{R_2/R_1}$ ☐ $-\sqrt{R_1/R_2}$ ☐ $\sqrt{R_2/R_1}$ ☐ $\sqrt{R_1/R_2}$ ☐ $-R_2/R_1$ ■ $-R_1/R_2$ ☐ R_2/R_1 ☐ R_1/R_2

Solution 1.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

For part (a), as stated in the hint, we can combine continuity equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} = 0 \text{ (since in steady state)} \tag{1}$$

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with Ohm's law, $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$*, to get*

$$\nabla \cdot \mathbf{E} = -\nabla \cdot (\mathbf{v} \times \mathbf{B}) \tag{2}$$

which gives the required expression via Gauss law:

$$\rho = \epsilon_0 \nabla \cdot \mathbf{E} = -\epsilon_0 \nabla \cdot (\mathbf{v} \times \mathbf{B}) \tag{3}$$

For part (b), we simply insert the relevant quantities:

$$\mathbf{v} \times \mathbf{B} = (\boldsymbol{\omega} \times \mathbf{r}) \times \mathbf{B} = \mathbf{r}(\mathbf{B} \cdot \boldsymbol{\omega}) - \boldsymbol{\omega}(\mathbf{B} \cdot \mathbf{r}) = -|\boldsymbol{\omega}||\mathbf{B}|(x\hat{x} + y\hat{y}) \tag{4}$$

hence

$$\rho = -\epsilon_0 \nabla \cdot (\mathbf{v} \times \mathbf{B}) = 2\epsilon_0 |\boldsymbol{\omega}||\mathbf{B}| \simeq 2 \times 10^{-11} \times 10 \times 5 = 10^{-9} \tag{5}$$

For part (c), use the hint:

$$\mathbf{J} \cdot \hat{r} = \sigma\,(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \hat{r} = 0 \quad \rightarrow \quad \mathbf{E} \cdot \hat{r} = -(\mathbf{v} \times \mathbf{B}) \cdot \hat{r} \tag{6}$$

hence

$$\frac{\partial V}{\partial r} = \hat{r} \cdot \nabla V = -\hat{r} \cdot \mathbf{E} = (\mathbf{v} \times \mathbf{B}) \cdot \hat{r} = [(\boldsymbol{\omega} \times \mathbf{r}) \times \mathbf{B}] \cdot \hat{r} \tag{7}$$

Since magnetic field is uniform and the angular velocity is constant, the only difference at the points A and B will be due to $\mathbf{r}$*. But they are collinear (and have opposite directions as origin is in between), so we simply have* $\mathbf{r}\big|_A = R_2\hat{s}$ *and* $\mathbf{r}\big|_B = -R_1\hat{s}$ *for some unit vector* $\hat{s}$*. Thus:*

$$\frac{\left.\dfrac{\partial V}{\partial r}\right|_{\text{at the point } B}}{\left.\dfrac{\partial V}{\partial r}\right|_{\text{at the point } A}} = \frac{[(\boldsymbol{\omega} \times (R_1\hat{s})) \times \mathbf{B}] \cdot \hat{s}}{[(\boldsymbol{\omega} \times (-R_2\hat{s})) \times \mathbf{B}] \cdot (-\hat{s})} = \frac{R_1}{R_2} \tag{8}$$

Question: 2: Poynting vector ..... *(31½ points)*
(Midterm One Question Expanded)

Please take $\epsilon_0 \simeq 10^{-11}$ *and* $\mu_0 \simeq 10^{-6}$ *for this question.*

Imagine a cylindrical conductive wire of radius $\pi^{-1/2}$ and length 2 in an otherwise empty space. Set up a cylindrical coordinate system such that the axis of the wire coincides with the positive z–axis, with one of its faces coinciding with the $z = 0$ plane. We will assume that the $z = 0$ face of the wire is held at the electric potential $V = V_0$ and the other face at $V = 3$. We are additionally told that the current density $\mathbf{J}$ satisfies $\mathbf{J}(z = 0, r, \theta) = 2\hat{z}$ and that the wire has resistivity linear in z, i.e. $\rho(z) = 1 + cz$ for some constant c.

(a) **(10½ points)** For $c = 1/2$, what is the Poynting vector $\mathbf{S}$ evaluated at $(r, \theta, z) = (\pi^{-1/2}, 0, \pi - 2)$?

☐ $-\pi^{-1/2}\hat{r}$ ☐ $\pi^{-1/2}\hat{r}$ ■ $-\pi^{1/2}\hat{r}$ ☐ $\pi^{1/2}\hat{r}$ ☐ $-\hat{r}$ ☐ $\hat{r}$ ☐ $-\hat{z}$ ☐ $\hat{z}$

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(b) **(10½ points)** For $c = 3/2$, what is the total work done per unit time by the electric & magnetic fields on the *part* of the material between $z = 0$ and $z = 1$?
Hint: you may relate resistance R to resistivity ρ via $dR = \rho d\ell/A$.

☐ 4 ☐ 5 ☐ 6 ■ 7 ☐ −4 ☐ −5 ☐ −6 ☐ −7

(c) **(10½ points)** For $c = 0$, what is the total work done per unit time on the electric & magnetic fields by the *part* of the material between $z = 0$ and $z = 1$?
Hint: work done by A on B is minus the work done by B on A.

☐ 4 ☐ 5 ☐ 6 ☐ 7 ■ −4 ☐ −5 ☐ −6 ☐ −7

Solution 2.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

By integrating the current density at $z = 0$, we can find the current at $z = 0$; but since there is no way for current to leak to some other system (as there is none), total current will be constant throughout the wire:

$$I = \int_{z=0\ \text{cross-section}} \mathbf{J} \cdot \mathbf{dA} = |\mathbf{J}| \int_{z=0\ \text{cross-section}} |\mathbf{dA}| = 2 \times \pi \times (\pi^{-1/2})^2 = 2 \tag{9}$$

From Faraday's law, this then leads to a constant magnetic field

$$\mathbf{B} = \frac{\mu_0 I}{2\pi r}\hat{\theta} = \frac{10^{-6} \times 2}{2 \times \pi \times \pi^{-1/2}}\hat{\theta} = \frac{10^{-6}}{\pi^{1/2}}\hat{\theta} \tag{10}$$

As the given resistivity depends only on z, we can consider infinitesimally thick cylindrical shells of length dz that will have resistance

$$dR = \rho A^{-1} dz = (1 + cz)(\pi \times (\pi^{-1/2})^2)^{-1} dz = (1 + cz)dz \tag{11}$$

Infinitesimal voltage drop on this shell is $dV = -I dR = -2(1 + cz)dz$; since $dV = (\nabla V) \cdot \mathbf{dr} = -\mathbf{E} \cdot \mathbf{dr}$, we conclude

$$\mathbf{E} = 2(1 + cz)\hat{z} \tag{12}$$

The Poynting vector is then

$$\mathbf{S} = \frac{1}{\mu}\mathbf{E} \times \mathbf{B} = 10^6(2(1 + cz)\hat{z}) \times \left(\frac{10^{-6}}{\pi^{1/2}}\hat{\theta}\right) = -\frac{2(1 + cz)}{\pi^{1/2}}\hat{r} \tag{13}$$

For part (a), we simply insert the given values, i.e. $c = 1/2$ and $(r, \theta, z) = (\pi^{-1/2}, 0, \pi - 2)$: $\mathbf{S} = -\sqrt{\pi}\hat{r}$.

*For parts (b) & (c), we first compute the total work done per unit time **by** the fields **on** the material,*

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hence we should integrate S *against the surface pointing inward towards the material, hence*

$$\int \mathbf{S}\cdot\mathbf{dA} = \int_{z=0\ face} \mathbf{S}\cdot\hat{z}|\mathbf{dA}| + \int_{z=1\ face} \mathbf{S}\cdot(-\hat{z})|\mathbf{dA}| + \int_{side\ face} \mathbf{S}\cdot(-\hat{r})|\mathbf{dA}|$$
$$= 0 + 0 + \int_{side\ face} \frac{2(1+cz)}{\pi^{1/2}}|\mathbf{dA}| = \int_0^1 \frac{2(1+cz)}{\pi^{1/2}}(2\times\pi\times\pi^{-1/2}\times dz) = 4+2c \tag{14}$$

For part (b), $c = 3/2$ *means that the answer is* 7.

For part (c), we need to be careful: the total work done per unit time ***on*** *the fields* ***by*** *the material is* $-4-2c$*, hence the answer is* -4.

One could have derived the answer for part (c) by realizing that $c = 0$ *means a constant resistivity* $\rho = 1$ *which gives* $R = \rho\frac{\ell}{A} = 1\frac{(1-0)}{\pi(\pi^{-1/2})^2} = 1$ *for the part between* $z = 0$ *and* $z = 1$*. But the total work done* ***by*** *the material is simply* $-RI^2 = -1\times 2^2 = -4$ *as derived above.*

Question: 3: Anisotropy in Linear Medium ..... *(21 points)*

(We partially solved this problem in class.)

Consider a linear homogeneous medium with permittivity and permeability tensors given as

$$\epsilon_{ij} \doteq \begin{pmatrix} 1+a & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\epsilon \quad , \quad \mu_{ij} \doteq \begin{pmatrix} 1 & 0 & 0 \\ 0 & (1+b)^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix}\mu \tag{15}$$

for some constants a, b, ϵ, and μ. Under the assumption that there is no free charges or free currents in the medium, we can derive the differential equation describing the z−component of the electric field in certain cases as

$$\begin{aligned} \left(\nabla^2 - \epsilon\mu\frac{\partial^2}{\partial t^2}\right)E_z(\mathbf{r},t) &= a\left[\alpha_1\frac{\partial^2 E_x}{\partial x\partial z} + \alpha_2\frac{\partial^2 E_z}{\partial x\partial z}\right] \qquad (\text{if } b=0) \\ \left(\nabla^2 - \epsilon\mu\frac{\partial^2}{\partial t^2}\right)E_z(\mathbf{r},t) &= b\left[\alpha_3\frac{\partial^2 B_x}{\partial t\partial y} + \alpha_4\frac{\partial^2 B_y}{\partial t\partial x}\right] \qquad (\text{if } a=0) \end{aligned} \tag{16}$$

(a) **(10½ points)** Consider the case $b = 0$; what is the correct tuple (α_1, α_2)?

☐ $(-1,-1)$ ■ $(-1,0)$ ☐ $(-1,1)$ ☐ $(0,-1)$ ☐ $(0,1)$ ☐ $(1,-1)$ ☐ $(1,0)$ ☐ $(1,1)$

(b) **(10½ points)** Consider the case $a = 0$; what is the correct tuple (α_3, α_4)?

☐ $(-1,-1)$ ☐ $(-1,0)$ ☐ $(-1,1)$ ■ $(0,-1)$ ☐ $(0,1)$ ☐ $(1,-1)$ ☐ $(1,0)$ ☐ $(1,1)$

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Solution 3.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

Let's start with the maxwell equations which is also given in the formulae sheet:

$$\nabla \cdot \mathbf{D} = \rho_f \ , \quad \nabla \cdot \mathbf{B} = 0 \ , \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \ , \quad \nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \tag{17}$$

We can write this in index notation using Einstein's summation convention (we also use the fact that there are no free charges or currents):

$$\partial_i D_i = 0 \ , \quad \partial_i B_i = 0 \ , \quad \epsilon_{ijk}\partial_j E_k = -\partial_t B_i \ , \quad \epsilon_{ijk}\partial_j H_k = \partial_t D_i \tag{18}$$

We now use the fact that the medium is linear, i.e. $D_i = \epsilon_{i\ell}E_\ell$ *and* $H_k = (\mu^{-1})_{k\ell}B_\ell$*:*

$$\partial_i(\epsilon_{i\ell}E_\ell) = 0 \ , \quad \partial_i B_i = 0 \ , \quad \epsilon_{ijk}\partial_j E_k = -\partial_t B_i \ , \quad \epsilon_{ijk}\partial_j \left((\mu^{-1})_{k\ell}B_\ell\right) = \partial_t(\epsilon_{i\ell}E_\ell) \tag{19}$$

Since we are given that a *and* b *are constants,* ϵ *and* μ *are simply constant tensors:*

$$\epsilon_{i\ell}\partial_i E_\ell = 0 \ , \quad \partial_i B_i = 0 \ , \quad \epsilon_{ijk}\partial_j E_k = -\partial_t B_i \ , \quad \epsilon_{ijk}(\mu^{-1})_{k\ell}\partial_j B_\ell = \epsilon_{i\ell}\partial_t E_\ell \tag{20}$$

We will now proceed as usual (as we have also done in class), which is also illustrated in the book for the isotropic case: take the curl of Faraday's law, insert Ampere's and Gauss's laws inside to get the wave equation. But before doing that, for the complete sake of illustration purposes, let's see what these equations look like in matrix notation:

$$\begin{pmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{pmatrix} \begin{pmatrix} 1+a & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} = 0 \ , \quad \begin{pmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{pmatrix} \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = 0 \tag{21a}$$

$$\begin{pmatrix} 0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x} \\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} = -\begin{pmatrix} \frac{\partial B_x}{\partial t} \\ \frac{\partial B_y}{\partial t} \\ \frac{\partial B_z}{\partial t} \end{pmatrix} \ , \quad \begin{pmatrix} 0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x} \\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1+b & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = \begin{pmatrix} 1+a & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{\partial E_x}{\partial t} \\ \frac{\partial E_y}{\partial t} \\ \frac{\partial E_z}{\partial t} \end{pmatrix} \tag{21b}$$

Clearly, one can work with purely compact vector notation

$$\nabla \times \mathbf{E} \ , \tag{22}$$

with index notation

$$\epsilon_{ijk}\partial_j E_k \ , \tag{23}$$

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or with matrix notation

$$\begin{pmatrix} 0 & -\frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & -\frac{\partial}{\partial x} \\ -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} . \tag{24}$$

Going back and forth between these notations is actually straightforward (and trivial once you get used to it); however, matrix notation is not that useful beyond second rank tensors and the compact vector notation is limited in all practical purposes to three dimensional space only (hence it is not that useful in special relativity). Therefore, we will proceed with index notation (as we have done in the class as well).

For part (a), (20) becomes

$$\partial_i E_i = -a\frac{\partial E_x}{\partial x} , \quad \partial_i B_i = 0 , \quad \epsilon_{ijk}\partial_j E_k = -\partial_t B_i , \quad \epsilon_{ijk}\partial_j B_k = \epsilon\mu\left(\frac{\partial E_i}{\partial t} + a\delta_{ix}\frac{\partial E_x}{\partial t}\right) \tag{25}$$

where we used

$$\epsilon_{ij} = \left(\delta_{ij} + a\delta_{ix}\delta_{jx}\right)\epsilon , \quad (\mu^{-1})_{ij} = \mu^{-1}\delta_{ij} \tag{26}$$

Let us now take the curl of Faraday's law and insert other equations as appropriate:

$$\begin{aligned} 0 =& \epsilon_{ijk}\partial_j E_k + \partial_t B_i \\ 0 =& \epsilon_{mni}\partial_n\left(\epsilon_{ijk}\partial_j E_k + \partial_t B_i\right) \\ 0 =& (\epsilon_{imn}\epsilon_{ijk})\partial_n\partial_j E_k + \partial_t(\epsilon_{mni}\partial_n B_i) \\ 0 =& \partial_m(\partial_k E_k) - (\partial_j\partial_j)E_m + \partial_t\left[\epsilon\mu\left(\frac{\partial E_m}{\partial t} + a\delta_{mx}\frac{\partial E_x}{\partial t}\right)\right] \\ 0 =& \partial_m\left[-a\frac{\partial E_x}{\partial x}\right] - \left[(\partial_j\partial_j) - \epsilon\mu\frac{\partial^2}{\partial t^2}\right]E_m + a\mu\epsilon\delta_{mx}\frac{\partial^2 E_x}{\partial t^2} \end{aligned} \tag{27}$$

hence

$$\left(\nabla^2 - \epsilon\mu\frac{\partial^2}{\partial t^2}\right)E_z(\mathbf{r}, t) = -a\frac{\partial^2 E_x}{\partial x\partial z} \tag{28}$$

For part (b), (20) becomes

$$\partial_i E_i = 0 , \quad \partial_i B_i = 0 , \quad \epsilon_{ijk}\partial_j E_k = -\partial_t B_i , \quad \epsilon_{ijk}\partial_j B_k - b\,\delta_{ix}\partial_z B_y + b\,\delta_{iz}\partial_x B_y = \epsilon\mu\partial_t E_i \tag{29}$$

where we used

$$\epsilon_{ij} = \delta_{ij}\epsilon , \quad (\mu^{-1})_{ij} = \mu^{-1}\left(\delta_{ij} + b\,\delta_{iy}\delta_{jy}\right) \tag{30}$$

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Let us now take the curl of Faraday's law and insert other equations as appropriate:

$$\begin{aligned}
0 =& \epsilon_{ijk}\partial_j E_k + \partial_t B_i \\
0 =& \epsilon_{mni}\partial_n \left(\epsilon_{ijk}\partial_j E_k + \partial_t B_i\right) \\
0 =& (\epsilon_{imn}\epsilon_{ijk})\partial_n\partial_j E_k + \partial_t(\epsilon_{mni}\partial_n B_i) \\
0 =& \partial_m(\partial_k E_k) - (\partial_j\partial_j)E_m + \partial_t\left[\epsilon\mu\partial_t E_m + b\,\delta_{mx}\partial_z B_y - b\,\delta_{mz}\partial_x B_y\right] \\
0 =& 0 - \left[(\partial_j\partial_j) - \epsilon\mu\frac{\partial^2}{\partial t^2}\right]E_m + b\,\delta_{mx}\frac{\partial^2 B_y}{\partial t\partial z} - b\,\delta_{mz}\frac{\partial^2 B_y}{\partial t\partial x}
\end{aligned} \tag{31}$$

hence

$$\left(\nabla^2 - \epsilon\mu\frac{\partial^2}{\partial t^2}\right)E_z(\mathbf{r},t) = -b\frac{\partial^2 B_y}{\partial t\partial x} \tag{32}$$

Question: 4: Attenuation of EM Waves in Conductors . . . . . *(21 points)*

Please take $\epsilon \simeq 5\epsilon_0 \simeq 5\times 10^{-11}$ and $\mu \simeq \mu_0 \simeq 10^{-6}$ for this question: this is a rather crude yet computationally helpful approximation.

Consider a metal block, of the shape right rectangular prism, of conductance σ, put on top of a level table. Choose the coordinate system such that the surface of the table is the $z = 0$ plane, and the facets of metal block are $z = 0$ (its facet touching the table), $z = 1$, $y = -1$, $y = 1$, $x = 0$, and $x = -1$.

Assume that we have waited sufficiently long so that the transient period has ended; in other words, $\mathbf{E}$–field satisfies the partial differential equation $\left(\nabla^2 - \mu\epsilon\frac{\partial^2}{\partial t^2} - \mu\sigma\frac{\partial}{\partial t}\right)\mathbf{E}(\mathbf{r},t) = 0$.

(a) **(10½ points)** For the real positive constants d & ω and the functions $k_\pm := \sqrt{d^{-2} \pm \epsilon\mu\omega^2}$, which option below can describe the real $\mathbf{E}$–field inside the metal for $|x| \ll 1$ and for a positive σ? *Hint: d denotes the **skin depth** and we expect the dielectric behavior as d goes to infinity.*

☐ $e^{-x/d}\cos(k_+x + \omega t)$ ☐ $e^{-x/d}\cos(k_+x - \omega t)$ ☐ $e^{-x/d}\cos(k_-x + \omega t)$ ☐ $e^{-x/d}\cos(k_-x - \omega t)$

■ $e^{x/d}\cos(k_+x + \omega t)$ ☐ $e^{x/d}\cos(k_+x - \omega t)$ ☐ $e^{x/d}\cos(k_-x + \omega t)$ ☐ $e^{x/d}\cos(k_-x - \omega t)$

(b) **(10½ points)** If we measure that a monochromatic blue light ($\omega \simeq 4\times 10^{15}$) inside the metal gets attenuated with a 100 nanometer skin depth ($d = 10^{-7}$), what would we calculate for the conductivity σ of this metal? *Hint: $\mathbf{E}$-field should satisfy the above partial differential equation.*

☐ 2×10^3 ☐ 5×10^3 ☐ 1×10^4 ☐ 1.5×10^4 ☐ 2×10^4 ☐ 5×10^4 ☐ 1×10^5 ■ 1.5×10^5

Solution 4.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

In part (a), we may first use the hint to deduce that all options with $\cos\left(\sqrt{d^{-2} - \epsilon\mu\omega^2}x \pm \omega t\right)$ are incorrect as they can not remain real in the limit $d \to \infty$. We further note that we are near the $x = 0$ facet of

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the metal as we are told that $|x| \ll 1$ (we are far away from the other facet, i.e. $x = -1$); therefore, the field should get attenuated as we decrease x (we are moving deeper in the metal), hence we need $e^{x/d} = e^{-|x|/d}$.

The two physical considerations above (correct dielectric limit and attenuation inside the metal) leaves us with two options:

$$\mathbf{E}_{\pm}(\mathbf{r}, t) = \mathbf{E}_0 e^{x/d} \cos\left(\sqrt{d^{-2} + \epsilon\mu\omega^2}x \pm \omega t\right) \tag{33}$$

We can further comment that we are describing a wave going into *the metal and getting attenuated, hence its wave vector* $\mathbf{k}$ *should have the direction* $-\hat{x}$*. This actually uniquely fixes the sign as a wave moving in the direction* $\mathbf{k}$ *has the form* $\cos(-\mathbf{k}\cdot\mathbf{r} + \omega t)$*. This is sufficient to solve the problem; however, let us not use this heuristic argument and determine the sign directly using the given differential equation:*

$$0 = \left(\nabla^2 - \mu\epsilon\frac{\partial^2}{\partial t^2} - \mu\sigma\frac{\partial}{\partial t}\right)\mathbf{E}(\mathbf{r}, t) \tag{34a}$$

$$= \left(\frac{\partial^2}{\partial x^2} - \mu\epsilon\frac{\partial^2}{\partial t^2} - \mu\sigma\frac{\partial}{\partial t}\right)\left[\mathbf{E}_0 e^{x/d}\cos\left(\sqrt{d^{-2} + \epsilon\mu\omega^2}x \pm \omega t\right)\right] \tag{34b}$$

$$= \pm\,\mu\omega\left(\sigma \mp \frac{2\sqrt{1 + d^2\epsilon\mu\omega^2}}{d^2\mu\omega}\right)\mathbf{E}_0 e^{x/d}\sin\left(\sqrt{d^{-2} + \epsilon\mu\omega^2}x \pm \omega t\right) \tag{34c}$$

Since all parameters are positive real numbers, the only way that this identity is satisfied is if

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{x/d}\cos\left(\sqrt{d^{-2} + \epsilon\mu\omega^2}x + \omega t\right) \tag{35a}$$

and if

$$\sigma = \frac{2\sqrt{1 + d^2\epsilon\mu\omega^2}}{d^2\mu\omega} \tag{35b}$$

Equations (35) make perfect sense! As $d \to 0$ (hence conductor becoming an ideal conductor), $\mathbf{E}$ field inside goes to zero (thanks to $e^{-|x|/d}$) and the conductivity σ goes to infinity. Conversely, $\sigma \to 0$ requires $d \to \infty$ and this reduces $\mathbf{E}$ to $\mathbf{E}_0\cos(kx + \omega t)$ for $k = \sqrt{\epsilon\mu}\omega$.

In part (b), we simply insert the numbers:

$$\begin{aligned}\sigma &= \frac{2\sqrt{1 + d^2\epsilon\mu\omega^2}}{d^2\mu\omega} \\ &= \frac{2\sqrt{1 + (10^{-7})^2(5\times10^{-11})(10^{-6})(4\times10^{15})^2}}{(10^{-7})^2(10^{-6})(4\times10^{15})} \\ &= \frac{2\sqrt{1+8}}{4\times10^{-5}} = 1.5\times10^5\end{aligned} \tag{36}$$

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It is intriguing to further comment on our result: for a cylindrical metal wire of this conductivity, of radius 1 mm and of length 1 meter, we have the resistance

$$R = \frac{\ell}{A\sigma} = \frac{1}{\pi(10^{-3})^2 \times 1.5 \times 10^5} \simeq 2\Omega \tag{37}$$

Considering that many electronic circuits work with resistors in kΩ range, such a wire can indeed be approximated as an ideal wire.

« « « Congratulations, you have made it to the end! » » »